

LLD Case Study 10: Multi-Car Elevator Control System

Target Patterns: State Pattern, Strategy Pattern, Command Pattern

Key Engineering Focus: Hardware movement states, SCAN / LOOK disk scheduling algorithms, internal/external hall call commands, and multi-elevator coordination.

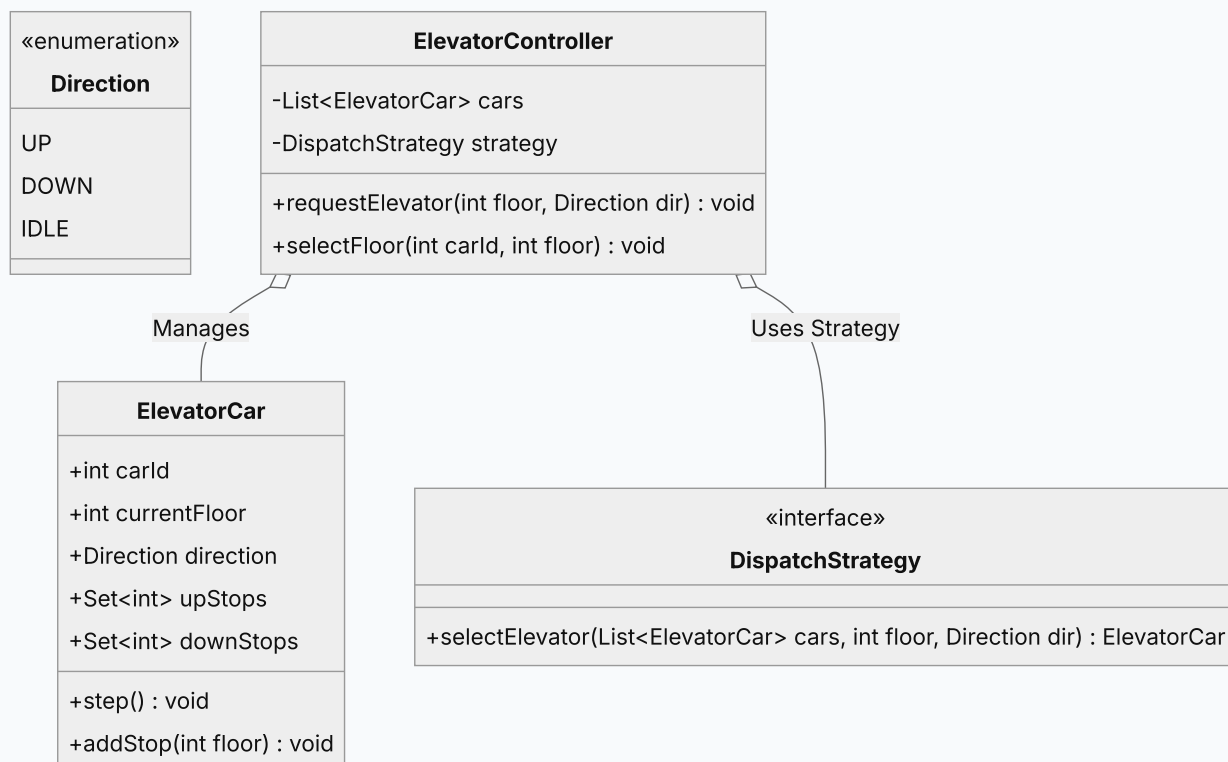
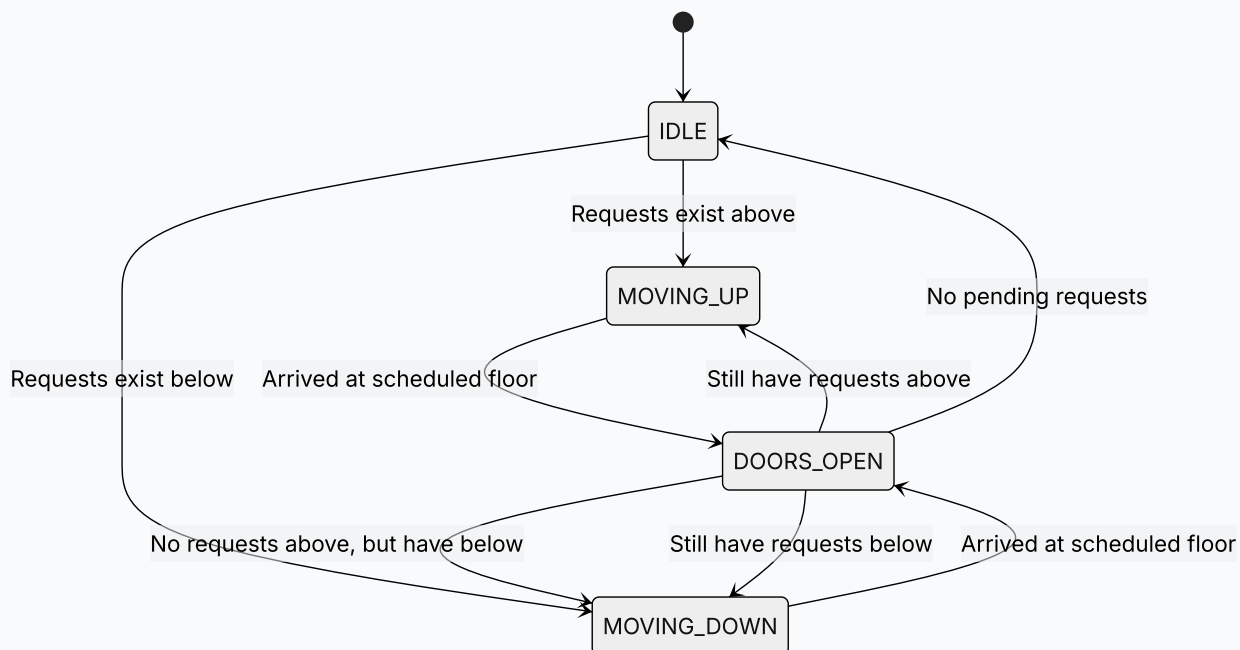
1. Problem Statement & Functional Requirements

Design an automated Elevator Controller governing a bank of M elevators across N floors.

Requirements:

- 1. Movement State Machine (State Pattern):** Elevators transition between `IDLE`, `MOVING_UP`, `MOVING_DOWN`, and `DOORS_OPEN`.
 - 2. Elevator Dispatch Policy (Strategy Pattern):** Route hall calls to the most optimal car using algorithms like **LOOK / SCAN** (continue in current direction servicing all pending stops before reversing).
 - 3. Command Architecture:** Requests come from external hall panels (`direction = UP/DOWN`) and internal car panels (`destination_floor`).
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2. Architecture & State Diagram



3. Production-Grade Python Implementation

